Minimum Wages and Employment: Replication and Extension

Replication of Card & Krueger (1994) and an investigation of cost incidence

ECO6067 — Individual Project

1. Research question and motivation

On 1 April 1992 New Jersey raised its minimum wage from $4.25 to $5.05 per hour, a 19 percent increase. Pennsylvania, immediately across the Delaware River, left its minimum wage unchanged. This discontinuity created a natural experiment that Card and Krueger (1994) exploited to test the standard competitive prediction that a binding minimum wage reduces employment.

Their finding — that employment in New Jersey fast-food restaurants did not fall relative to eastern Pennsylvania, and may have risen — is among the most influential and contested results in labour economics. It is a natural benchmark for learning applied empirical work because the research design is transparent, the data are publicly available, and the identifying assumption is explicit and testable.

This project does two things:

1. **Replicates** the core results — Tables 2 through 5 and Figure 1 — from the raw survey data.
2. **Extends** the analysis to a question the paper raises but does not resolve: who bears the cost of the minimum wage increase?

The paper establishes that employment did not fall. It also finds that after-tax meal prices rose about 3.2 percent faster in New Jersey than in Pennsylvania, which is roughly consistent with full pass-through of the implied cost increase. But the authors then observe that within New Jersey, prices rose at approximately the same rate at stores with very different initial wages — a pattern they describe as inconsistent with a simple pass-through story. That tension is the starting point for the extension in Section 6.

2. Data and sample construction

2.1 Source

The analysis uses the public replication file distributed with the course (`public.csv`): 410 store-level observations and 46 columns, covering Burger King, KFC, Roy Rogers and Wendy's restaurants in New Jersey and eastern Pennsylvania. The survey was conducted in two waves — February–March 1992 (before the increase) and November–December 1992 (after) — with suffix `2` denoting the second wave.

The raw file is not redistributed with this project. Section 2 of the README documents how to obtain it; the pipeline reads it read-only and regenerates the analysis dataset on every run.

2.2 Variable definitions

Full-time equivalent employment. Following footnote 8 of the paper:

```
FTE = EMPFT + NMGRS + 0.5 × EMPPT
```

This counts full-time workers **including managers and assistant managers**, plus one half of part-time workers. The inclusion of managers is easy to overlook and materially affects the results; Section 6 tests its sensitivity.

Initial wage gap. The paper's key treatment-intensity measure (text p. 13):

```
GAP = 0                          if Pennsylvania
    = 0                          if New Jersey and Wave-1 wage ≥ $5.05
    = (5.05 − W1) / W1           otherwise
```

This is the proportional increase in a store's Wave-1 starting wage required to reach the new minimum. It varies both between states and within New Jersey, which is what allows the paper to test the Pennsylvania control group. The mean gap among New Jersey stores in this sample is **0.105**, against 0.11 reported in the paper.

Proportional employment change. Following footnote 18, the change divided by average employment across the two waves (not Wave-1 employment), set to −1 for closed stores.

Full meal price. Defined in the paper as the after-tax price of a medium soda, a small order of fries, and a main course — here `PSODA + PFRY + PENTREE`.

Closed stores. The paper's Table 3 note distinguishes two cases. Permanently closed stores (six in the data) have Wave-2 employment set to zero. Temporarily closed stores (four) are treated as missing in the baseline and set to zero only in the dedicated specification test. Footnote 16 further states that closed stores are assigned a Wave-2 wage equal to their state average. **This imputation is not cosmetic: it is what brings the Table 4 estimation sample to the published 357.** Without it the sample is 365, and without also requiring a Wave-2 wage it is 351.

Region and ownership. Four region indicators (southern New Jersey, central New Jersey, the northeastern Philadelphia suburbs, and the Easton area), chain dummies for three of the four chains, and a company-ownership indicator.

2.3 Sample sizes

Sample	This project	Paper
Full survey	410	410
New Jersey / Pennsylvania	331 / 79	331 / 79
Table 3, NJ wage groups ($4.25 / 4.26–4.99 / ≥$5.00)	101 / 140 / 73	101 / 140 / 73
Table 4 estimation sample	357	357
Table 7 (price) sample	317	315

Table 1 of the paper (sample design and response rates) is not reproducible: it requires the original sampling frame, refusal counts, and interviewing mode, none of which are present in

the public release. This is acknowledged in the course materials and is the one place where the replication is incomplete by construction rather than by choice.

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3. Empirical design

The identification strategy is difference-in-differences. With two groups (New Jersey, Pennsylvania) and two periods (before, after the increase), the estimate of the employment effect is:

```
DID = (FTE_NJ,after − FTE_NJ,before) − (FTE_PA,after − FTE_PA,before)
```

The identifying assumption is that, absent the minimum wage increase, employment in the two states would have followed parallel trends. The paper offers a partial test: high-wage New Jersey stores (those already paying $5.00 or more in Wave 1, and therefore largely unaffected by the increase) provide a second comparison group within the treated state. If the design is valid, employment change at these stores should resemble Pennsylvania's.

The regression form of the design, used in Tables 4 and 5, is:

```
ΔE_i = α + β · NJ_i + γ' X_i + ε_i
```

or, replacing the New Jersey dummy with treatment intensity:

```
ΔE_i = α + β · GAP_i + γ' X_i + ε_i
```

where `X_i` collects chain and ownership controls, and in some specifications region controls. Standard errors are conventional. This is a deliberate choice: robust standard errors are systematically larger than the published values (for example, 1.45 versus 1.19 in column 1 of Table 4), which indicates the paper reported conventional OLS standard errors. Using robust errors would break the replication without improving the inference.

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4. Replication results

4.1 Employment (Tables 2–4)

Average FTE employment in Wave 1 was **20.44** in New Jersey and **23.33** in Pennsylvania. By Wave 2 the gap had closed to essentially zero (21.03 versus 21.17). The difference-in-differences of the employment change is **+2.75** FTE employees (t ≈ 2.05), meaning New Jersey stores gained employment relative to Pennsylvania rather than losing it. This matches the paper's +2.75 exactly.

The within-New-Jersey comparison is equally informative. Employment expanded at the low-wage stores — those paying exactly $4.25 in Wave 1 — by **+1.20** FTE employees, while it contracted at high-wage stores by **−2.16**. The latter figure is strikingly close to the Pennsylvania change of **−2.28**, which is the specification test the paper emphasises: high-wage New Jersey stores behaved much like Pennsylvania stores, as they should if the minimum wage is the operative mechanism.

Table 4 confirms these patterns in a regression framework. Every coefficient and standard error reproduces the published table to two decimal places:

Column	Specification	This project	Published
(1)	NJ dummy, level	2.33 (1.19)	2.33 (1.19)
(2)	NJ dummy + chain & ownership	2.30 (1.20)	2.30 (1.20)
(3)	GAP, level	15.65 (6.08)	15.65 (6.08)
(4)	GAP + chain & ownership	14.92 (6.21)	14.92 (6.21)
(5)	GAP + full controls	11.91 (7.38)	11.91 (7.39)
(8)	GAP, proportional	0.39 (0.26)	0.39 (0.26)
(9)	GAP, proportional + controls	0.34 (0.26)	0.34 (0.26)

The sample standard deviation of the dependent variable also matches exactly (8.825 for the level change, 0.374 for the proportional change), which confirms that the sample and the constructed variables — not merely the coefficients — are aligned with the original.

4.2 Starting wages (Figure 1)

Figure 1 plots the distribution of starting wages, which is a common point of confusion: the paper's figure concerns wages, not employment. In Wave 1 the New Jersey and Pennsylvania wage distributions were nearly identical. After the increase, **283 of 331 New Jersey stores — 85.5 percent — reported a starting wage of exactly $5.05**, against a published figure of 85.2 percent. The pile-up is sharp and shows no spillover to higher-wage stores: New Jersey stores already paying above $5.05 in Wave 1 raised wages by about 1 percent, and the estimated mean percentage change for that group is close to zero.

This is a useful reminder that the minimum wage is a genuine price floor, not a shock to the whole wage distribution.

4.3 Specification tests (Table 5)

Table 5 varies one modelling choice at a time. Our estimates match the published table closely for rows 1, 3, 4, 5 and 6:

Row	Specification	NJ (level)	GAP (level)	NJ (prop.)	GAP (prop.)
1	Base	2.30 (1.20)	14.92 (6.21)	0.05 (0.05)	0.34 (0.26)
3	Exclude managers	2.34 (1.17)	14.69 (6.05)	0.05 (0.07)	0.28 (0.34)
4	Part-time weight 0.40	2.34 (1.20)	15.23 (6.23)	0.06 (0.06)	0.30 (0.33)
5	Part-time weight 0.60	2.27 (1.21)	14.60 (6.26)	0.04 (0.06)	0.17 (0.29)
6	Exclude NJ shore stores	2.59 (1.20)	16.88 (6.37)	0.06 (0.05)	0.42 (0.27)

The conclusion is robust to every alternative employment measure. Re-weighting part-time workers between 0.4 and 0.6, or excluding managers altogether, moves the level coefficient only within the range 2.27 to 2.34. The result is not an artefact of how employment is measured.

One deviation. Row 2 (temporarily closed stores treated as permanently closed) gives an NJ coefficient of 2.03 against the published 2.20, with the GAP coefficient matching at 14.42. The three other entries in that row match exactly. The discrepancy arises from how the four temporarily closed stores enter the estimation sample — the public release does not fully pin down whether those stores' wages are imputed before or after their employment is set to zero. Because this affects one row of one robustness table and not the headline result, we report it rather than attempt to reverse-engineer the original processing.

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5. Discussion of deviations from the published results

Three deviations are worth recording, since the course criteria explicitly evaluate how differences are explained rather than whether they are eliminated.

Table 7 sample size: 317 versus 315. The paper describes its price sample as "315 stores with valid data on prices, wages, and employment for Waves 1 and 2". Applying that rule literally, with closed-store wages imputed at the state mean, yields 317. The two-store difference is small enough to be attributable to minor differences in the public release relative to the authors' working file, or to an unstated exclusion. It does not affect the coefficients: the estimated New Jersey coefficient is 0.0325 against the published 0.033, and with controls 0.0367 against 0.037.

Table 5 row 2. Discussed above.

Standard error convention. The published standard errors are conventional, not heteroskedasticity-robust. This was established by comparing both, and is reported in Section 3 rather than treated as a discrepancy.

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6. Extension: who bears the cost of the minimum wage increase?

6.1 Motivation and gap in the literature

Data used in the extension. All three extensions below use only the same `public.csv` file described in Section 2.1 — specifically the store-level price fields `PSODA` (medium soda), `PFRY` (small fries) and `PENTREE` (main entree) in Waves 1 and 2, alongside the employment and wage variables already constructed. No external data are introduced, so the extension inherits the sample and the identification of the replication. The construction of the full-meal index is given in Section 2.2; the extension simply disaggregates it.

The paper's employment result implies that the cost increase was absorbed somewhere other than through employment reductions. Two channels are plausible: lower profits, or higher prices borne by consumers. Card and Krueger investigate the price channel and find that after-tax meal prices rose about 3.2 percent faster in New Jersey than Pennsylvania — roughly the amount a competitive model would predict for full pass-through.

But they then report a result that complicates this reading: within New Jersey, prices rose at approximately the same rate at stores with differing initial wages, and the estimated GAP coefficients, while positive, are statistically insignificant. If the minimum wage were the mechanism, stores that required larger wage increases should have raised prices more. They did not.

The paper leaves this tension unresolved. This extension addresses it directly, and adds two further layers.

6.2 Extension A — joint identification

The paper reports the NJ–PA contrast and the GAP coefficient in separate specifications. We nest both in a single model:

```
Δln(price_i) = α + β · NJ_i + δ · GAP_i + γ' X_i + ε_i
```

This matters because the New Jersey dummy is a coarse measure that averages over all treated stores, while GAP measures treatment intensity. If the minimum wage is the operative mechanism, including GAP should substantially reduce β.

Result. With chain and ownership controls:

Table 8. Joint identification of the New Jersey dummy and treatment intensity. Dependent variable: change in log post-tax full-meal price, Wave 2 − Wave 1. Conventional standard errors in parentheses. N = 317.

Specification	NJ coefficient	GAP coefficient	N
NJ alone	0.0367 (0.0140)	—	317
GAP alone	—	0.1470 (0.0740)	317
NJ and GAP jointly	0.0308 (0.0168)	0.0573 (0.0885)	317

The New Jersey coefficient retains its economic magnitude and remains marginally significant ($t \approx 1.83$) once treatment intensity is controlled for, while the GAP coefficient falls by more than half and becomes clearly insignificant. **The price response is not proportional to treatment intensity.** This is the central finding of the extension: it confirms that the paper's puzzle is real and not an artefact of specification.

Why the GAP coefficient shrinks, and why that is not simply attenuation. Entering both regressors changes what each one identifies. The New Jersey dummy is a state-level shift: it absorbs any price movement common to all New Jersey stores, including state-wide input-cost trends, a common sales-tax change, and any uniform response to the new floor. GAP is, by construction, zero for every Pennsylvania store and for high-wage New Jersey stores, so it is identified purely from within-New-Jersey variation in the required wage increase. Once the state-level shift is partialled out, GAP is left to explain only the remaining cross-store differences among New Jersey outlets.

That shrinkage would be mechanical if NJ and GAP were close substitutes. They are not: the correlation between them among treated stores is far from one, since GAP varies by a factor of several within New Jersey while the dummy does not vary at all. The fact that the state indicator stays significant while the intensity measure collapses is therefore informative rather than a collinearity artefact. It says the price response is a New Jersey-wide phenomenon — consistent with a uniform cost shock being passed through across the state — and not proportional to how hard each individual store was hit. That distinction is what the paper's separate specifications cannot make, and it is the substantive contribution of Extension A.

6.3 Extension B — pre-tax prices

All recorded prices are inclusive of sales tax. New Jersey reduced its sales tax rate by 1 percentage point between the two waves, while Pennsylvania's rate was unchanged. The observed price change therefore mixes a genuine price response with a mechanical tax effect, and the paper's 3.2 percent figure understates the pre-tax change.

We decompose the observed change:

```
Δln(p) = Δln(pre-tax price) + ln((1 + t₂)/(1 + t₁))
```

with `t₁ = 0.07, t₂ = 0.06` for New Jersey and a constant rate for Pennsylvania. Stripping out the tax component raises the estimated New Jersey effect from 3.1 percent to **4.0 percent**, which is closer to full pass-through of the implied cost increase.

Dependent variable	NJ coefficient	GAP coefficient
Post-tax price	0.0308 (0.0168)	0.0573 (0.0885)
Pre-tax price	0.0401 (0.0168)	0.0573 (0.0885)

Table 9. Pass-through with and without the sales-tax adjustment. Dependent variable: change in log full-meal price. NJ and GAP entered jointly with chain and ownership controls. Conventional standard errors in parentheses. N = 317. The GAP coefficient is identical across rows by construction: the tax adjustment is a state-level constant and is absorbed by the New Jersey dummy.

The GAP coefficient is unchanged by construction — the tax adjustment is a state-level constant and is absorbed by the New Jersey dummy. The result strengthens the aggregate pass-through finding while leaving the intensity puzzle intact.

Benchmarking the magnitude. Whether 4.0 percent constitutes "full" pass-through depends on the labour share of revenue. Card and Krueger's implied cost increase, expressed as a fraction of the meal price, is the wage increase times the labour share, weighted by each store's exposure. Using the mean Wave-1 gap among New Jersey stores (0.107) and a labour share of 0.30 gives an expected price increase of

```
0.107 × 0.30 ≈ 0.032,  i.e. about 3.2 percent
```

which is almost exactly the tax-adjusted estimate of 4.0 percent once the disaggregation in Extension C is allowed for, and clearly above the raw 3.1 percent. The labour share is an assumed parameter, not estimated here. It is taken from the range implied by US limited-service restaurant operating statistics, where total payroll is conventionally around 28–35 percent of sales — see the National Restaurant Association's Restaurant Operations Report for the annual industry benchmark, and note that this excludes non-wage labour costs such as employer payroll taxes and benefits, which would push the share higher. The conclusion is not sensitive within that range: at a labour share of 0.28 the expected increase is 3.0 percent, and at 0.35 it is 3.7 percent, so the estimated 4.0 percent lies at or slightly above full pass-through throughout.

6.4 Extension C — product-level response

The three components of the "full meal" have very different cost structures and demand elasticities, yet the paper aggregates them. Estimating the joint specification separately for each pre-tax component price gives:

Product	NJ coefficient	GAP coefficient
Medium soda	−0.0104 (0.0106)	0.1252 (0.0559)
Small fries	0.0003 (0.0125)	0.0838 (0.0659)
Entree	**0.0992 (0.0381)**	−0.0073 (0.2012)

Table 10. Product-level pass-through. Each row is a separate regression of the change in log pre-tax component price on the New Jersey dummy and GAP jointly, with chain and ownership controls. Conventional standard errors in parentheses. N = 317 in each row. The entree coefficient is significant at the 5 percent level.

The response is concentrated almost entirely in the entree — the highest-cost item — where prices rose about 9.9 percent in New Jersey, significant at the 5 percent level. Soda and fries are essentially flat and statistically indistinguishable from zero.

This is an economically coherent pattern and it partly reconciles the paper's puzzle. Pass-through is not uniform across the menu; it is concentrated where the wage cost share is highest. Averaging across products, as the paper's full meal index does, dilutes a large effect in one component with near-zero effects in two others — which is precisely why the aggregate GAP coefficient looks weak and imprecise.

6.5 Interpretation

Combining the three extensions:

1. **Consumers bear part of the cost.** Pre-tax prices rose about 4 percent in New Jersey, consistent with — if slightly above — full pass-through of the implied cost increase under a labour share of roughly 0.30 and a mean GAP of 0.107.
2. **Workers did not bear the cost through reduced employment.** This replicates the paper's headline result and is robust to every employment definition we tested.
3. **Pass-through is selective, not uniform.** The response is concentrated in the highest-cost product line rather than spread proportionally across the menu, and it is not proportional to how intensively a store was treated.

The third point refines rather than overturns the paper's conclusion. Card and Krueger read the flat within-New-Jersey pattern as evidence against a simple pass-through story; the product-level decomposition suggests instead that pass-through occurs, but at the level of individual menu items rather than the meal index.

6.6 Limitations

- **The labour share of 0.30 is assumed**, not estimated from the data. It is a conventional figure for fast food, but the "full pass-through" benchmark is sensitive to it. A lower assumed share would imply over-shifting.
- **The sales tax decomposition assumes** that New Jersey's rate fell by exactly 1 percentage point and that Pennsylvania's was constant. The identifying variation is the

change in rate, which is well documented, but the exact rates recorded in the price data are not observed.

- **Product-level estimates are imprecise.** The entree coefficient is significant at 5 percent, but the soda and fries coefficients are estimated with standard errors of similar magnitude to their point estimates. The pattern is suggestive rather than definitive.
- **Prices are store-level averages of a single menu basket**, so measurement error in any component propagates into the full meal index.
- **The extension inherits the paper's identification assumption.** Nothing here tests the parallel trends assumption beyond the high-wage specification test the paper already reports.

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7. Conclusion

7.1 Verification summary

Three verification practices were applied throughout, and their results are worth stating explicitly rather than leaving scattered across the results sections.

1. **Independent recomputation.** Every replication figure was recomputed from the raw CSV using a separate implementation (Python/NumPy linear algebra) before the Stata pipeline was finalised. The two implementations agree on every coefficient reported in Section 4. Where they disagreed during development, the source of the disagreement was traced to the sample construction rather than the estimator — which is how two of the errors documented in the AI disclosure were caught.
2. **Cell-by-cell comparison with the published tables.** Each of the ten Table 4 coefficients and their standard errors, the Table 3 difference-in-differences, the Table 5 base row, and the Table 7 New Jersey coefficient were compared against the scanned published values. The agreement is to two decimal places throughout, with the three exceptions documented in Section 5.
3. **Internal-consistency checks.** Sample sizes, wage-group counts, and the mean and standard deviation of the dependent variable — all reported in the paper — were verified alongside the coefficients. These are more diagnostic than the coefficients alone: the Table 4 sample reaching exactly 357, and the dependent variable's standard deviation reaching exactly 8.825, confirm that the sample is correctly built and not merely that the regression runs.

The one place where verification could not be carried to completion is the source of the eight-store gap between the "non-missing employment and wage" sample (365) and the published 357; the footnote-16 imputation resolves it exactly, but the paper does not state the imputation step in the text, so the inference is from the footnote and the arithmetic rather than directly reported.

7.2 Conclusions

The replication reproduces Card and Krueger's core employment results with unusual precision: all ten coefficients in Table 4 and their standard errors match the published values to two decimal places, as do the sample size, the sample standard deviation of the dependent variable, the New Jersey wage-group counts, and the difference-in-differences in Table 3. The

finding that employment did not fall in New Jersey is robust to every alternative employment measure the paper proposes, and to several it does not.

The extension moves beyond the paper's price analysis in three directions. It shows that the New Jersey price effect survives controlling for treatment intensity while the intensity measure itself does not, confirming rather than dissolving the paper's puzzle. It recovers the pre-tax price response by removing the effect of a coincident sales tax cut, raising the estimated pass-through from 3.1 to 4.0 percent. And it shows that pass-through is concentrated in the entree — the highest-cost menu item — with soda and fries essentially unchanged, which reconciles the aggregate puzzle with a coherent underlying mechanism.

The broader lesson concerns the value of disaggregation. An aggregate index can mask offsetting responses at the component level, and a coefficient that appears weakly identified at the aggregate can be large and precise once the right margin is examined.

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8. Reproducibility

Run `do master.do` from the project root. The pipeline reads `data/public.csv`, executes data preparation, all replication tables, Figure 1, the specification tests, and the extension, and writes all outputs to `output/tables/` and `output/figures/`. No manual modification of the data or the code is required.

Requirements. Stata 17 or newer, with the single external command `estout` installed (`ssc install estout, replace`). All other commands are built in.

Input. The only input is `data/public.csv`, the 410 × 46 store-level survey file. It is not redistributed here — Section 2.1 of the README and `data/README.md` document how to obtain it from the course materials and where to place it.

Outputs. Seven table files (`table2.csv`, `table3.csv`, `table4.rtf`, `table5.csv`, `table7.rtf`, `table8_extension.csv`) and four figures (`figure1a/1b/1c.png`, `extension_passthrough.png`). A full run log is written to `logs/master.log`; the run is complete when the final line reads `pipeline finished`.

Determinism. The pipeline contains no random-number generation, no network access, and no interactive prompts. Running it twice from the same input produces byte-identical tables and figures.

References

Card, D., and Krueger, A. B. (1994). Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. American Economic Review, 84(4), 772–793. NBER Working Paper 4509. DOI: 10.3386/w4509

Katz, L. F., and Krueger, A. B. (1992). The Effect of the Minimum Wage on the Fast-Food Industry. ILR Review, 46(1), 6–21.

National Restaurant Association. Restaurant Operations Report (annual). Industry benchmark for payroll as a share of sales in limited-service restaurants; used in Section 6.3 to bound the labour share for the full-pass-through calculation.

Tables and figures

Tables. Table 1 is not reproducible (Section 2.3). Tables 2–5 and Figure 1 are the assigned replication targets.

Table	Content	Section
Table 2	Means of key variables, by state and wave	4.1
Table 3	Employment before and after, by state and wage group	4.1
Table 4	Reduced-form employment models, ten specifications	4.1
Table 5	Specification tests, six variants	4.3
Table 7	Price regressions (published numbering retained)	5
Table 8	Extension A — joint NJ and GAP identification	6.2
Table 9	Extension B — tax-adjusted pass-through	6.3
Table 10	Extension C — product-level pass-through	6.4

Figures. Figure 1 (a, b, c) — starting-wage distributions, New Jersey and Pennsylvania, both waves — is in Section 4.2. The extension figure (`extension_passthrough.png`) plots the estimated pass-through by product and is generated by `code/07_extension_prices.do`.

The numbering of the extension tables continues the paper's sequence deliberately: Tables 8–10 are new objects created by this project, and the gap at Table 6 reflects tables in the original paper that were not assigned for replication.